

Muhammad Usman

Full Stack Engineer

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Education

Air University

Bachelor of Science in Computer Science

Sep 2019 - June 2023

Islamabad, Pakistan

Experience

Tekrowe Digital — Digital Product Studio building SaaS and AI-powered platforms
Full Stack Engineer

July 2023 - Present

Gulberg, Islamabad

- Full-Stack Engineer with ~3 years of experience building and scaling SaaS platforms and real-time systems. Skilled in backend architecture, API design, and performance optimization with MERN and NestJS. Built and integrated AI-driven features such as chatbots and voice agents into production applications. Uses AI-assisted tooling to ship faster and improve development efficiency. Focused on delivering scalable, high-impact systems.

Projects

Nymbi - Conversational AI Platform (Healthcare)

Full-Stack Engineer | React.js, NestJS, PostgreSQL, LangChain, SIP, LiveKit

Conversational AI platform delivering chatbot and voice bot solutions for healthcare domain.

- Built production chatbot and AI voice agent for healthcare customer automation, handling appointment booking etc flows using SIP and LiveKit.
- Designed voice workflows integrating speech-to-text, LangChain, and backend APIs — enabling natural, context-aware conversations at scale.
- Developed intelligent appointment booking flows with third-party API integrations across chat and voice channels.

Syntera Automation - Healthcare Workflow Platform

Full-Stack Engineer | React.js, NestJS, PostgreSQL, Sockets

Healthcare workflow platform used by nursing staff to track resident care.

- Designed and shipped a speech-to-text nursing documentation system, cutting manual data entry time by ~50% across care staff workflows.
- Implemented real-time patient monitoring via WebSockets, enabling caregivers to track resident status without page refresh.
- Delivered frontend and backend in a regulated healthcare environment with strict data integrity requirements.

PerfAi - API Security & Performance SaaS

Full-Stack Engineer | Vue.js, NestJS, MongoDB, SSE

SaaS platform focused on API security, testing, and performance optimization.

- Owned core backend and frontend for a high-traffic SaaS product — optimizations contributed to 20% improvement in overall application performance.
- Built real-time data streaming via Server-Sent Events, reducing sync delays by ~30% and improving responsiveness for concurrent users.
- Integrated GitHub, Slack, and Jira — drove 20% increase in user engagement through workflow automation.
- Collaborated with design team to reduce onboarding friction and improve task completion rates.

Technical Skills

Frontend: React.js, Next.js, Vue.js, JavaScript (ES6+), Tailwind CSS

Backend: Node.js, Express.js, NestJS, REST APIs, WebSockets, SSE

Database: MongoDB, PostgreSQL, Redis

AI & Voice: LangChain, OpenAI API, SIP, LiveKit, Speech-to-Text, Vector Databases (Pinecone), Embeddings, RAG Pipelines

Auth & Security: JWT, OAuth, RBAC

Tools: Git, Docker, GitHub Actions, AWS (EC2, S3, Lambda), Cursor IDE

Personal Projects

Return Ai - E-Commerce Review Intelligence System

Full-Stack Engineer | RAG Pipeline, Vector Database, OpenAI Embeddings, LangChain

- Built an end-to-end RAG pipeline over e-commerce product reviews, enabling semantic search and context-aware Q&A on large review datasets.
- Implemented vector embeddings and similarity search using Pinecone to retrieve relevant review context before LLM generation
- Designed the retrieval pipeline to reduce hallucination and ground AI responses in actual customer sentiment data.

Forenex

Full-Stack Engineer | MERN

Data-driven workforce wellbeing and productivity platform designed to help teams detect burnout early, retain top talent, and optimize workload distribution.

- Built a real-time SaaS platform that analyzes task activity, engagement patterns, and workload signals to surface early indicators of employee burnout and disengagement.
- Designed task management workflows not just for execution, but to provide actionable insights into team health, productivity trends, and work balance.
- Architected the platform using the MERN stack, enabling scalable real-time data processing and rapid feature iteration.
- Implemented role-based access control and secure authentication, allowing managers and teams to access insights while maintaining data privacy.
- Developed a real-time notification and alerting system to proactively flag overload risks, missed recovery periods, and workflow bottlenecks.

Built scalable RESTful APIs to support future expansion into analytics dashboards, AI-driven recommendations, and HR system integrations.